

A. General foundation-model pattern

Heterogeneous inputs are embedded into a central foundation and adapted to many downstream tasks.

Inputs

Images and video

Text and documents

Audio and speech

Code and structured data



Foundation model

shared learned substrate



Downstream use cases

Question answering

Coding assistance

Summarization

Planning and retrieval

B. Bounded AI-assisted architecture study

The center connects architectural representations to tools and experiments under evidence and architect judgment.

Architecture inputs

Workload traces
and benchmarks

Specs, constraints,
and budgets

RTL, IP, configs,
and libraries

PDK, EDA, and simulator
assumptions

Verification and
evidence logs



AI-assisted architecture study

representations + tools
experiments + evidence
architect judgment



Study outputs

ISA and extension
exploration

Microarchitecture
proposals

Accelerator and SoC options

RTL and testbench fragments

DSE and verification
packages

AI-assisted studies connect representations, tools, experiments, evidence, and architectural judgment.